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Hardware

Nicla Vision

Tutorials

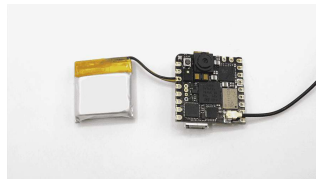
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Video Live Streaming with OpenMV and Nicla Vision

This tutorial will show you how to use the Nicla Vision to stream real-time video to your local network.

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Nicla Vision streamer overview

Overview

In this tutorial you will use the Arduino® Nicla Vision to capture and stream a live video feed to your local network. For that, you will use the onboard camera sensor and the Wi-Fi® connectivity. For this task you will write a MicroPython script and run it on the Nicla Vision with the help of the OpenMV IDE.

Goals

Learn how to use the OpenMV IDE to run MicroPython on Nicla Vision

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Overview

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Learn how to use MicroPython to capture images and stream them to the network

Required Hardware and Software

Nicla Vision

Micro USB cable (either USB-A to Micro USB or USB-C® to Micro USB)

OpenMV IDE 4.0.14+

Access to a Wi-Fi® local network (2.4Ghz only)

Nicla Vision and the OpenMV IDE

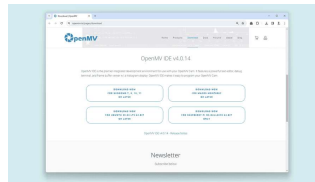
The OpenMV IDE was built for Machine Vision applications. It is meant to provide an Arduino like experience for simple computer vision tasks using a camera sensor. OpenMV comes with its own firmware that is built on MicroPython. Among other hardware, it supports the Nicla Vision board. OpenMV allows you to easily preview the camera stream and visually inspect color ranges to define thresholds for your machine vision scripts. [Here](#) you can read more about the OpenMV IDE.

Instructions

Configuring the Development Environment

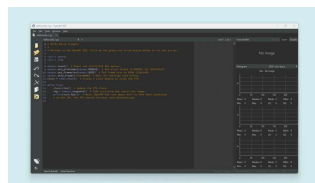
Before you can start programming micropython scripts for the Nicla

Open the [OpenMV](#) download page in your browser, download the latest version available for your operating system, and follow the instructions of the installer.



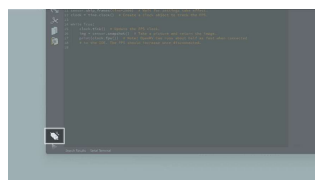
OpenMV Download Page

Open the OpenMV IDE and connect the Nicla Vision to your computer via the USB cable if you have not done so yet.



The OpenMV IDE after starting it

Click on the "connect" symbol at the bottom of the left toolbar.

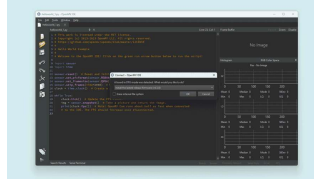


Click the connect button to attach the Nicla Vision to the OpenMV IDE

If your Nicla Vision does not have the latest firmware, a pop-up will ask you to install it. Your board will enter in DFU mode and its green LED will start fading.

Select

firmware on the Nicla Vision. You can leave the option of erasing the internal file system unselected and click **OK**.



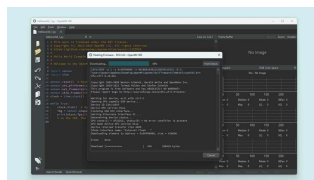
Install the latest version of the OpenMV firmware

Nicla Vision's green LED will start flashing while the OpenMV firmware is being uploaded to the board. A loading bar will start showing you the flashing progress.

Wait until the green LED stops flashing and fading. You will see a message saying

DFU firmware update complete!

when the process is done.



Installing firmware on Nicla Vision board in OpenMV

The board will start flashing its blue LED when it is ready to be connected. After confirming the completion dialog, the Nicla Vision should already be connected to the OpenMV IDE, otherwise, click the "connect" button (plug symbol) once again (the blue blinking should stop).



When the Nicla Vision is
successfully connected a
green play button appears

Live Streaming

In this section you will learn how the MicroPython script works in order to capture live video and stream it to your local network. Live video streaming across local or internet networks is one of the most used applications for connected cameras.

Some application examples are:

Monitor industrial processes,
from analog gauges, food-
industry processes, to CNC
machinery or 3D printers

Local video surveillance

Robotics, from SLAM navigation
with AGVs to computer vision-
powered robotics arms

The Example Code

This tutorial's main objective is to leverage the Nicla Vision to stream live video reachable from inside the network using your favorite web browser.

The example code we are going to use can also be found in OpenMV by navigating to **File > Examples > WiFi > mjpeg_streamer**.



In the newer versions of the OpenMV IDE you must connect the board to your computer so see the examples.

You can copy and paste the script below to test this tutorial application.

```
1 import sensor
2 import time
3 import network
4 import socket
5
6 SSID = "*****" # Network
7 KEY = "*****" # Password
8 HOST = "" # Use first interface
9 PORT = 8080 # Arbitrary
10
11 # Init wlan module and connect
12 wlan = network.WLAN(network.STA_IF)
13 wlan.active(True)
14 wlan.connect(SSID, KEY)
15
16 while not wlan.isconnected():
17     print('Trying to connect...')
18     time.sleep_ms(1000)
19
20 # We should have a valid connection
21 print("WiFi Connected ", wlan.local_ip())
22
23 # Create server socket
24 s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
25 s.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
26
27 # Bind and listen
28 s.bind([HOST, PORT])
```

Now, let's briefly explain the main code sections of the example script from above.

1. Prepare the Script

Create a new script by clicking the "New File" button in the toolbar on the left side. Delete the example program that initially appears and import the required modules:

```
1 import sensor # Import the sensor module
2 import time   # Import the time module
3 import network # Import the network module
4 import socket  # Import the socket module
```

Define your WiFi® credentials so the Nicla Vision can connect to the internet.

```
1 SSID = "*****" # Netw
2 KEY = "*****" # Ne
3 HOST = "" # Use first ava
4 PORT = 8080 # Arbitrary n
```

Set the `SSID` and `KEY` variables with your Wi-Fi® network credentials respectively.

The `HOST` variable must be empty so the socket uses the DHCP IP assigned to the Nicla Vision for the connection.

You can use `8080` as the `PORT` by default. Check with your network administrator if the port is closed or reserved so you know which one you can use for this application.

The code section below searches for a WiFi® network with the supplied credentials and starts a connection process to it. Also, it creates a socket server for video streaming.

```
1 # Init wlan module and co
2 wlan = network.WLAN(network
3 wlan.active(True)
4 wlan.connect(SSID, KEY)
5
6 while not wlan.isconnecte
7     print('Trying to conn
8     time.sleep_ms(1000)
9
10 # We should have a valid
11 print("WiFi Connected ",
12
13 # Create server socket
14 s = socket.socket(socket.
15 s.setsockopt(socket.SOL_S
16
17 # Bind and listen
18 s.bind([HOST, PORT])
19 s.listen(5)
20
21 # Set server socket to bl
22 s.setblocking(True)
```

3. Preparing the Sensor

```
1 # Camera setup and initial
2 sensor.reset() # Resets th
3 sensor.set_framesize(sensc
4 sensor.set_pixformat(sensc
```

The most relevant functions in this snippet are `set_pixformat` and `set_framesize`. The camera that comes with the Nicla Vision supports RGB 565 images. Therefore you need to set it via the `sensor.RGB565` parameter.

The resolution of the camera needs to be set to a supported format both by the sensor and the algorithm. `QVGA` is a good trade-off between performance and resolution so you will use this resolution in the tutorial.

The `start_streaming()` function waits for a client to get connected to the socket server, prints the client address and sends it to the headers preparing the connection for live streaming.

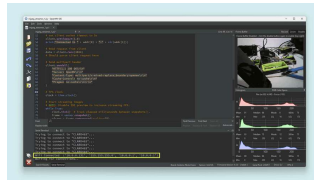
Eventually, it captures images continuously and sends them to the client compressed and in the right format.

```
1 def start_streaming(s):
2     print("Waiting for client")
3     client, addr = s.accept()
4     # set client socket options
5     client.settimeout(5)
6     print("Connected to " + addr)
7
8     # Read request from client
9     data = client.recv(1024)
10    # Should parse client request
11
12    # Send multipart headers
13    client.sendall(
14        "HTTP/1.1 200 OK\r\n"
15        "Server: OpenMV\r\n"
16        "Content-Type: image/jpeg\r\n"
17        "Cache-Control: no-cache\r\n"
18        "Pragma: no-cache\r\n"
19    )
20
21    # FPS clock
22    clock = time.clock()
23
24    # Start streaming images
25    # NOTE: Disable IDE serial monitor
26    while True:
27        clock.tick() # 1 second
28        frame = sensor.snapshot()
```

5. Uploading the Script

Let's program the board with the complete script and test if everything works as expected. Copy the whole [example code](#) and paste it into the

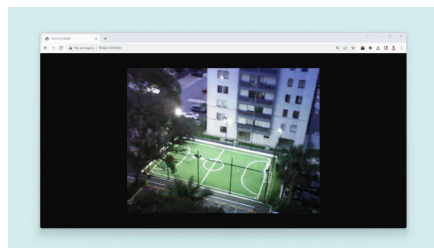
Open the OpenMV Serial Monitor by clicking on *Serial Terminal* in the lower left corner. Click the *Play* button at the bottom of the left toolbar. See the board connection progress in the terminal, once connected, the *host address* will be printed out.



Running the streaming example

The Nicla Vision IP address is the **first one** printed, `10.0.0.131` in this case.

To watch the live stream, open your favorite web browser and enter the Nicla Vision IP address followed by the port, in this case, `10.0.0.131:8080`.

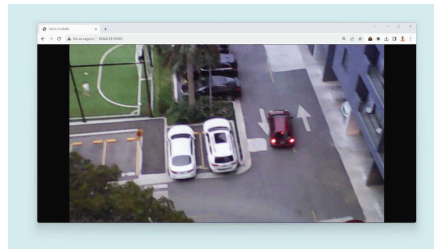


You can play with the camera resolution by changing it in the `set_framesize` as follows:

```
1 sensor.set_framesize(sensor.VGA)
2 sensor.set_framesize(sensor.VGA)
```

You can see all the supported resolutions [here](#) to suit your application.

the streaming fps.



Conclusion

In this tutorial, you learned how to use the OpenMV IDE to develop MicroPython scripts that then run on the Nicla Vision. You also learned how to connect it to the network via WiFi®. Last but not least, you learned how to configure the Nicla Vision to be used as a surveillance camera thanks to OpenMV.

Next Steps

Using a secure domain and a free DNS provider like [CloudFlare](#) you can access your live stream from outside your local network wherever you are.

Familiarize yourself with the OpenMV IDE. There are many other features that were not mentioned in this tutorial (e.g. Machine Learning built-in functions).

Learn more about the **Nicla Vision** on its [User Manual](#).

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